

3	(a) (i)	any two from 0.3(0) s, 0.9(0) s, 1.50 s (<i>allow 2.1 s etc.</i>)	B1	[1]
	(ii)	<i>either</i> $v = \omega x$ and $\omega = 2\pi/T$	C1	
		$v = (2\pi/1.2) \times 1.5 \times 10^{-2}$	M1	
		$= 0.079 \text{ m s}^{-1}$	A0	[2]
	<i>or</i>	gradient drawn clearly at a correct position	(C1)	
		working clear	(M1)	
		to give $(0.08 \pm 0.01) \text{ m s}^{-1}$	(A0)	

Page 3	Mark Scheme	Syllabus	Paper
	GCE A LEVEL – October/November 2013	9702	42

- (b) (i) sketch: curve from $(\pm 1.5, 0)$ passing through $(0, 25)$ M1
reasonable shape (*curved with both intersections between*
 $y = 12.0 \rightarrow 13.0$) A1 [2]
- (ii) at max. amplitude potential energy is total energy B1
total energy = 4.0 mJ B1 [2]